# Machine Learning Workflow Documentation

## 1. Data Collection

In this step, data is gathered from a reliable source. For this project, a student performance dataset was created containing Study Hours, Attendance, Assignments, and Marks. The collected data is relevant to predicting student marks. Good-quality data is important because it helps the model learn accurate patterns.

## 2. Data Cleaning

Data cleaning ensures that the dataset is free from errors, duplicates, and missing values. Clean data improves model performance and reliability. In this project, the dataset was checked to ensure all values were valid and properly formatted. This step helps avoid incorrect predictions.

## 3. Feature Selection

Feature selection involves choosing the input variables that influence the target variable. In this project, Study Hours, Attendance, and Assignments were selected as features. Marks were chosen as the target variable. Selecting relevant features helps improve model accuracy and reduces unnecessary complexity.

## 4. Train-Test Split

The dataset was divided into training data and testing data. Around 80% of the data was used for training and 20% for testing. The training data helps the model learn patterns, while the testing data evaluates performance. This ensures that the model can work well on new unseen data.

## 5. Model Training

A Linear Regression model was used for training. The model learned the relationship between the selected features and student marks. During training, it calculated coefficients and an intercept value. These values form the equation used for making predictions.

## 6. Testing

After training, the model was tested using the test dataset. Predictions were generated for the testing records and compared with actual marks. This step helps determine whether the model has learned meaningful patterns. Testing is essential for measuring reliability.

## 7. Evaluation

The model was evaluated using metrics such as Mean Absolute Error (MAE) and $R^2$ Score. MAE measures the average prediction error, while $R^2$ Score shows how well the model explains the data. Good evaluation scores indicate that the model performs effectively.

## 8. Prediction

Once the model was trained and evaluated, it was used to predict marks for new students. Predictions were made based on Study Hours, Attendance, and Assignments. This demonstrates the practical use of Machine Learning. The final goal is to make accurate predictions for future data.